

Hospital Patient Analytics

1. Project Overview

This project focuses on analyzing hospital patient data from **2019 to 2024** using a Power BI interactive dashboard. The dashboard provides insights into **patient volume, recovery trends, disease distribution, blood types, test results, and demographics**, supporting data-driven decision-making in hospital management.

2. Objectives

The primary goals of the dashboard were:

- To visualize patient **admissions and trends** over multiple years
- To track **average recovery days, average billing, and total patient count**
- To analyze **age-wise disease prevalence**
- To study **blood group distributions** to optimize blood bank management
- To understand the breakdown of **test results** and **gender distribution**
- To filter patient insights based on **admission type** (Elective, Emergency, Urgent)

3. Key Performance Indicators (KPIs)

✓ **Average Recovery Days: 15.50 Days**

Shows patients recover on average within **two weeks**, indicating stable hospital care routines.

✓ **Total Number of Patients: 55K**

Reflects consistent patient intake throughout the observed years.

✓ **Average Bill: 25.54K**

Indicates average treatment and hospitalization cost.

4. Patient Trends Over the Years (2019–2024)

The line chart shows:

Year	Patient Count
2019	7.3K
2020	11.2K

2021 10.8K

2022 10.9K

2023 10.9K

2024 3.8K

Insights

- **2020–2023 were peak years** with consistent patient volume >10K.
- **2024 saw a significant drop** (3.8K), indicating possible external events, resource constraints, or system changes.
- 2019 marked the lowest initial intake before growth.

5. Admission Type Insights

The dashboard segments patients into:

- **Elective**
- **Emergency**
- **Urgent**

Observation:

A balanced distribution shows that the hospital handles a mix of routine, critical, and semi-critical cases, reflecting broad capability.

6. Disease by Age Analysis

The age–disease matrix highlights:

Ages 13–17

- Lower incidence of severe diseases
- Most common: **Asthma, Diabetes, Obesity**

Ages 18–22 (High-risk group)

Sharp increase in **all major diseases**:

- Arthritis: 128–130 cases
- Asthma: ~130
- Cancer: ~125
- Diabetes: ~130–134
- Hypertension: up to 140
- Obesity: 118–132

Insights

- Young adults show **early onset of chronic diseases**, requiring immediate preventive programs.
- Hypertension and diabetes are the dominant conditions.

7. Blood Type Distribution

Treemap shows near-equal distribution:

- A-, A+, B+, AB+, AB-, B-, O+, O-
- Ranges from **6.80K to 6.90K**

Insight

- Blood group availability is uniform, aiding efficient blood bank inventory planning.

8. Test Type Breakdown

Three categories:

- **Normal**
- **Abnormal**
- **Inconclusive**

Insight

A noticeable amount of **abnormal** and **inconclusive** results indicates:

- Need for improved diagnostic accuracy

- Possible equipment calibration checks
- Laboratory process auditing

9. Gender Distribution

Shows a **balanced split between male and female patients**, indicating neutral patient demographics with no major bias.

10. Overall Findings

Positive Observations

- Stable recovery duration across years
- Strong patient intake during 2020–2023
- Healthy distribution of blood groups
- Clear segmentation by test results and admission types

Areas of Concern

- Sharp decline in patient count in 2024
- High chronic disease presence among 18–22 year-olds
- Inconclusive test results may signal operational issues

11. Recommendations

1. Investigate 2024 Decline

Evaluate:

- Hospital policy changes
- Capacity limitations
- Competition or external health events

2. Strengthen Preventive Healthcare for Youth

Launch:

- Awareness campaigns
- Lifestyle counseling
- Annual screening drive

3. Improve Diagnostic Procedures

- Upgrade equipment
- Enhance lab personnel training
- Audit test processing steps

4. Reduce Recovery Duration Further

Introduce:

- Improved physiotherapy programs
- Enhanced discharge planning
- Monitoring tools for post-operation care

5. Continue Monitoring Admission Types

Supports resource allocation in emergency vs elective units.

12. Conclusion

The Hospital Patient Analytics Dashboard delivers valuable insights into patient flows, disease patterns, resource allocation, and operational performance.

It highlights both **strengths** and **gaps**, helping hospital leaders make informed decisions aimed at enhancing patient care, operational efficiency, and long-term planning.